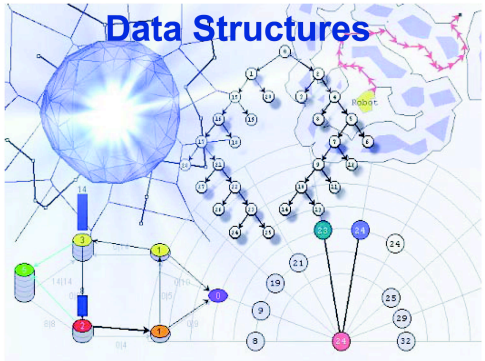
BBM 201
DATA STRUCTURES

Lecture 8:
Balanced Trees



Balanced Search Trees

- The height of a binary search tree is sensitive to the order of insertions and deletions.
  - The height of a binary search tree is between $\lceil \log_2(N+1) \rceil$ and N.
  - So, the worst case behavior of some BST operations are O(N).
- There are various search trees that can retain their balance despite insertions and deletions.
  - AVL Trees
  - 2-3 Trees
  - 2-3-4 Trees
  - Red-Black Trees
- In these height balance search trees, the run time complexity of insertion, deletion and retrieval operations will be $O(\log_2 N)$ at worst case.

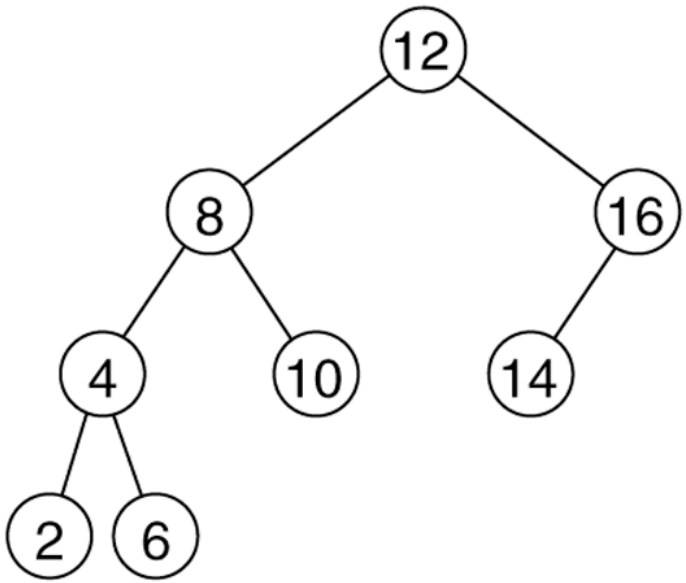
AVL Trees

- An AVL tree is a binary search tree with a *balance* condition.
- AVL is named for its inventors: **A**del'son-**V**el'skii and **L**andis
- AVL tree *approximates* the ideal tree (completely balanced tree).
- AVL Tree maintains a height close to the minimum.

Definition:

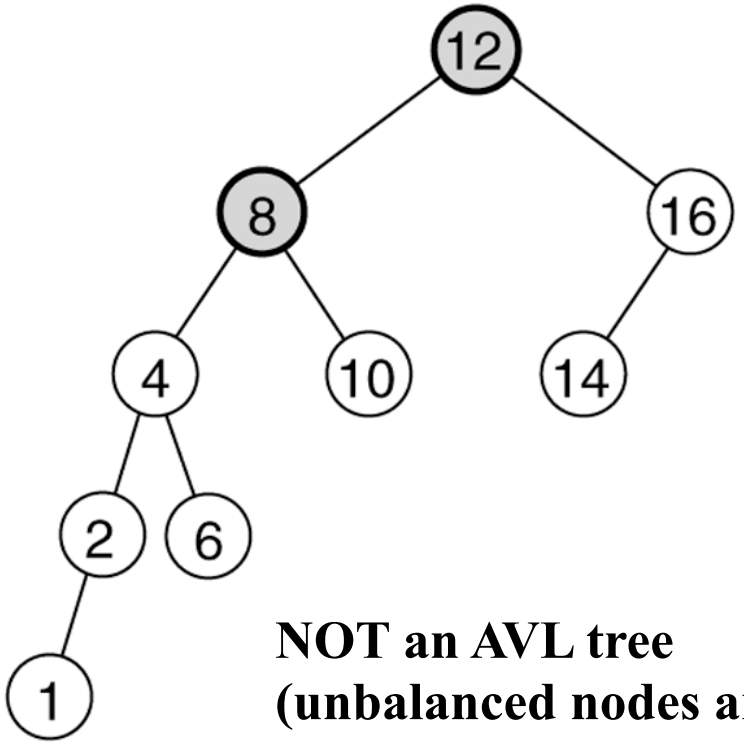
An AVL tree is a binary search tree such that for any node in the tree, the height of the left and right subtrees can differ by at most 1.

AVL Tree Examples



an AVL tree

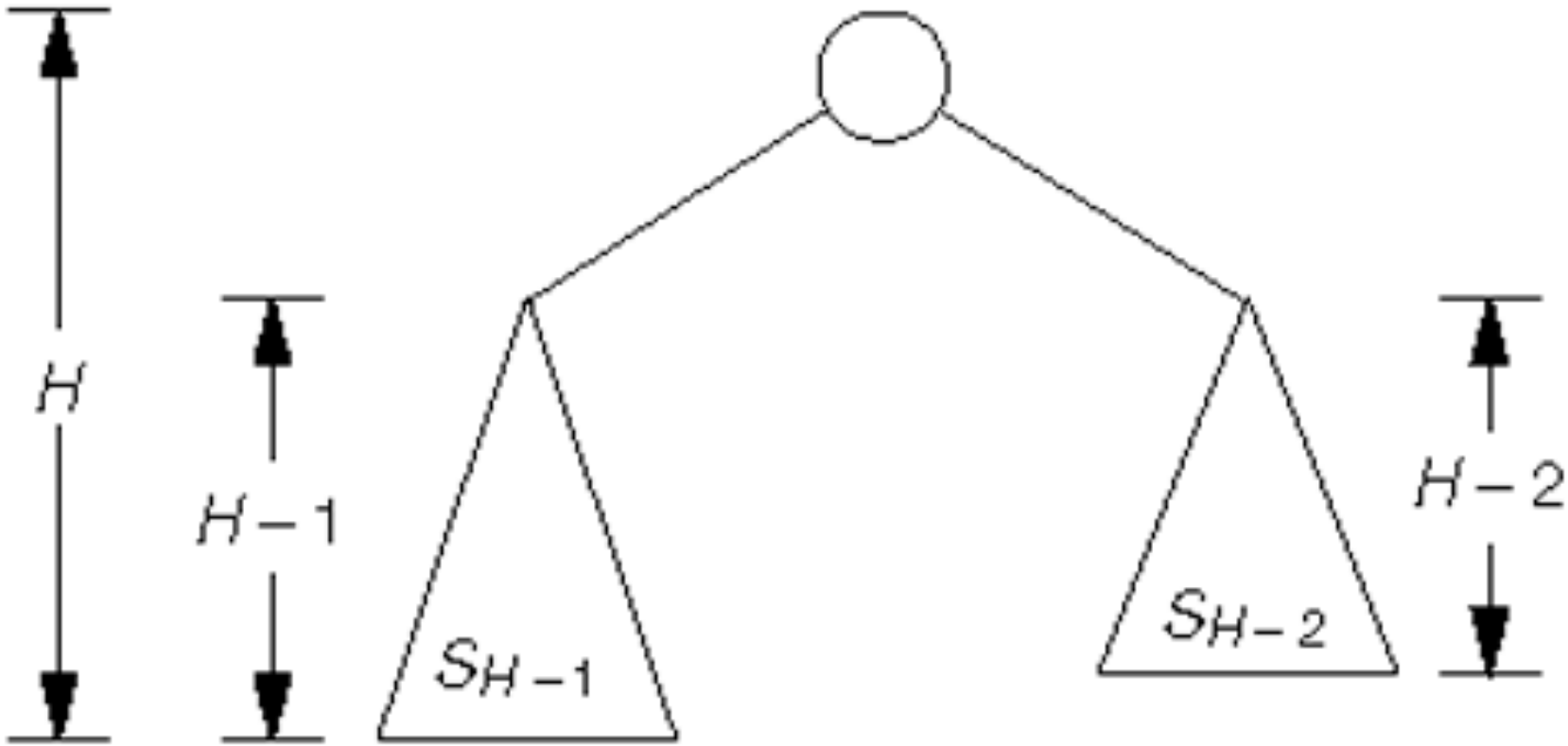
(a)



NOT an AVL tree (unbalanced nodes are darkened)

(b)

A Minimum Tree of height *H*



AVL Tree Properties

- The depth of a typical node in an AVL tree is very close to the optimal $log_2 N$.
- Consequently, all searching operations in an AVL tree have logarithmic worst-case bounds.
- An update (insert or delete) in an AVL tree could destroy the balance.

  ➔ It must then be rebalanced before the operation can be considered complete.
- *After an insertion, only nodes that are on the path from the insertion point to the root can have their balances altered.*

- Let us call the node that must be rebalanced x.
- Since any node has at most two children, and a height imbalance requires that x's two subtrees' heights differ by two,
- It is easy to see that a violation might occur in four cases:

1. An insertion into the left subtree of the left child of x
2. An insertion into the right subtree of the left child of x
3. An insertion into the left subtree of the right child of x
4. An insertion into the right subtree of the right child of x

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The first case, in which the insertion occurs on the "outside" (i.e., left–left or right– right), is fixed by a **single rotation** of the tree.

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4. An insertion into the right subtree of the right child of x

The second case, in which the insertion occurs on the "inside"
(i.e., left–right or right–left)
is handled by the slightly more complex **double rotation.**

Single Rotation

- A single rotation switches the roles of the parent and child while maintaining the search order.
- We rotate between a node and its child (left or right).
  - Child becomes parent.
  - Parent becomes right child in case 1, left child in case 4.
- The result is a binary search tree that satisfies the AVL property.

Single right rotation to fix case 1

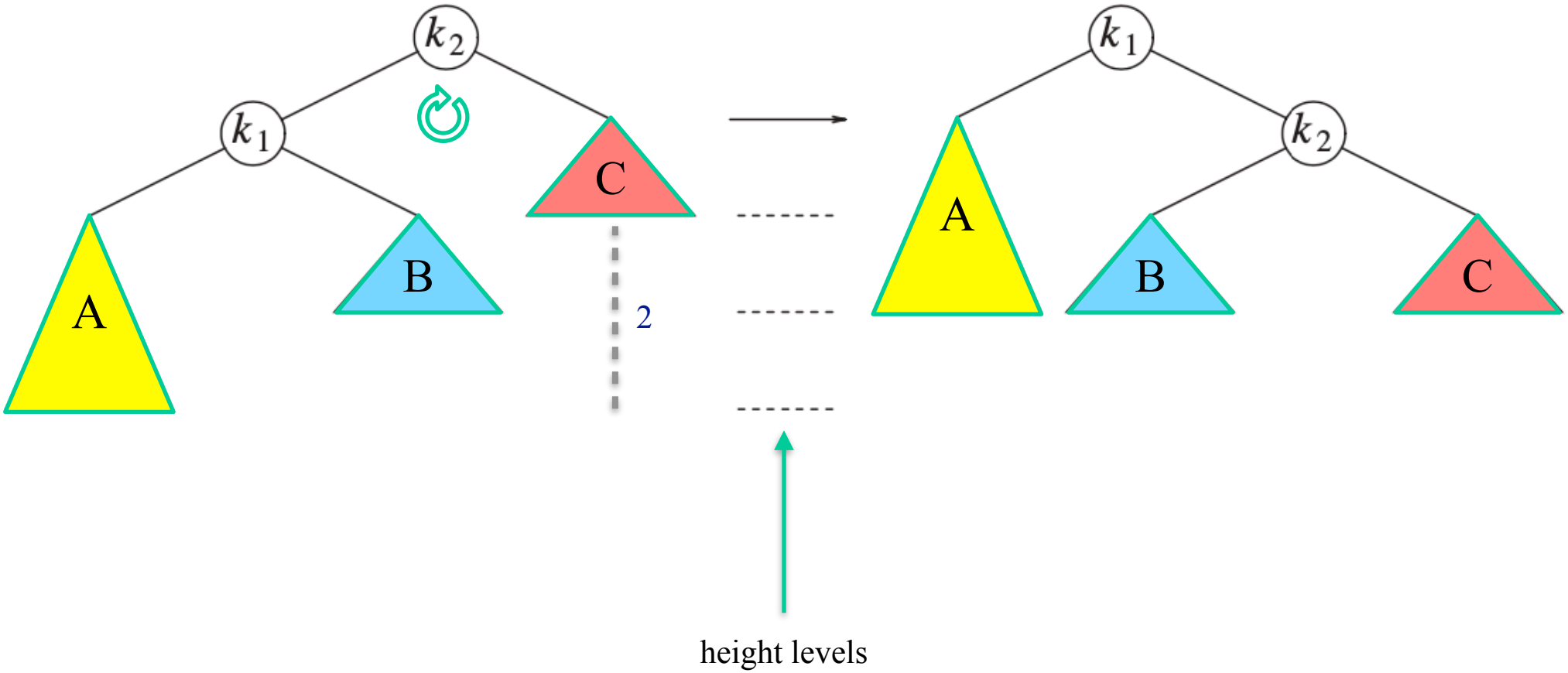
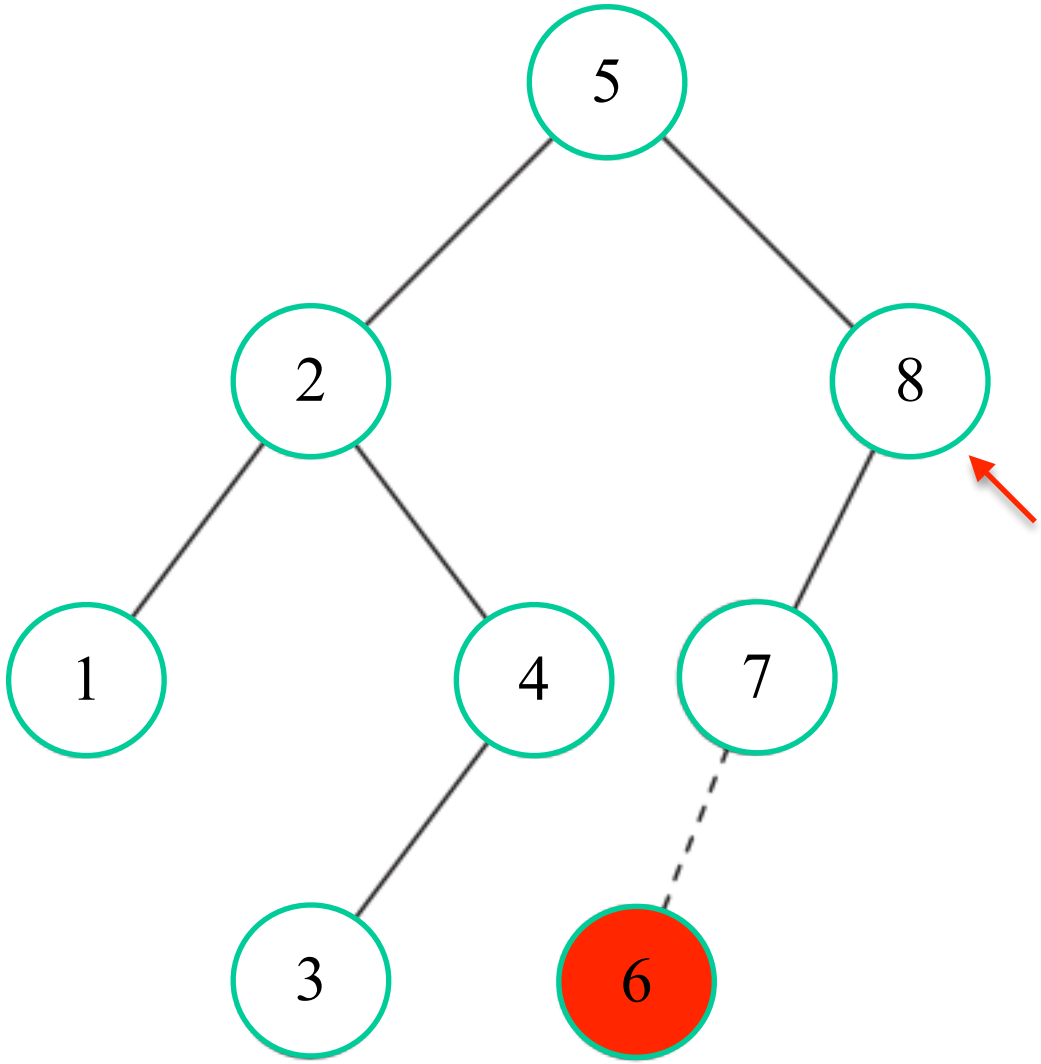


Image reference: Data Structures and Algorithm Analysis in C++. 4th Edition. Mark Allen Weiss, Pearson, 2014

AVL property violated by insertion of 6



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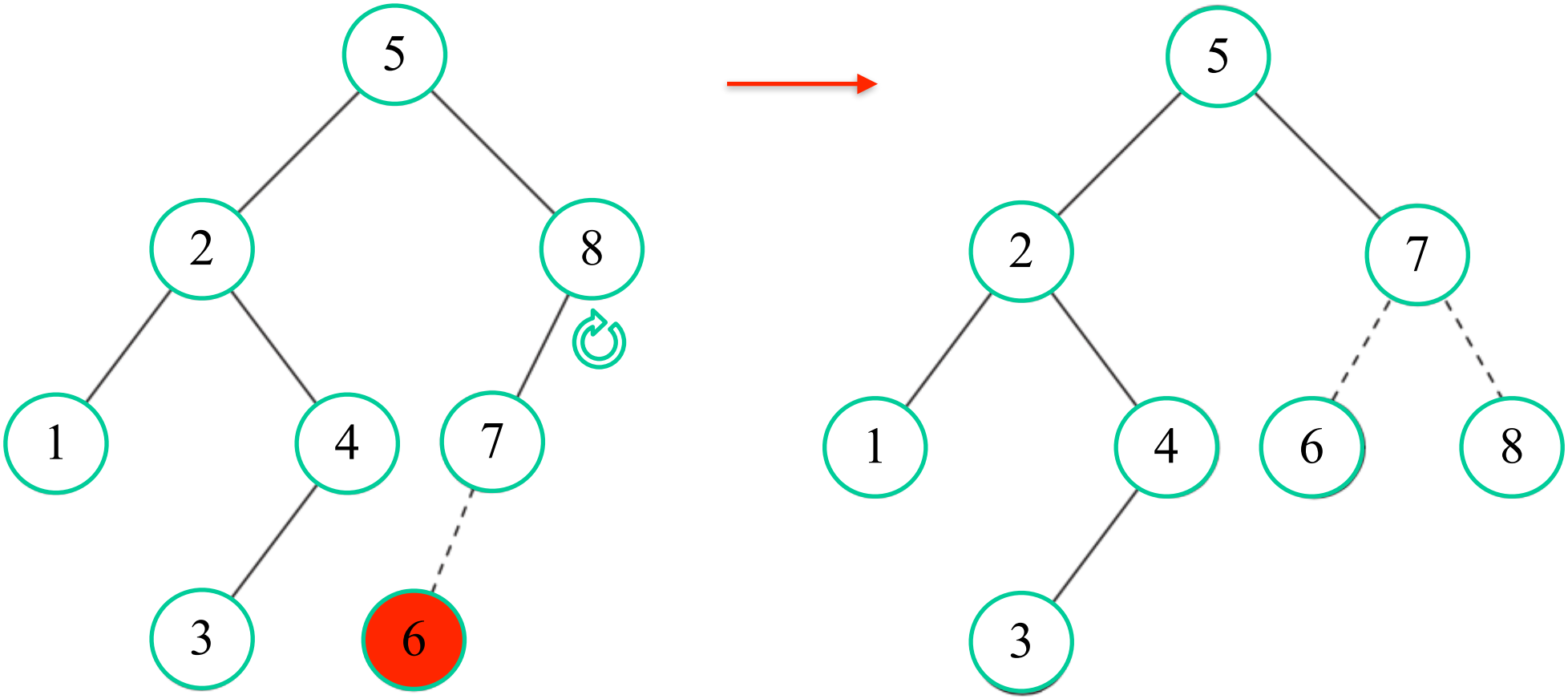

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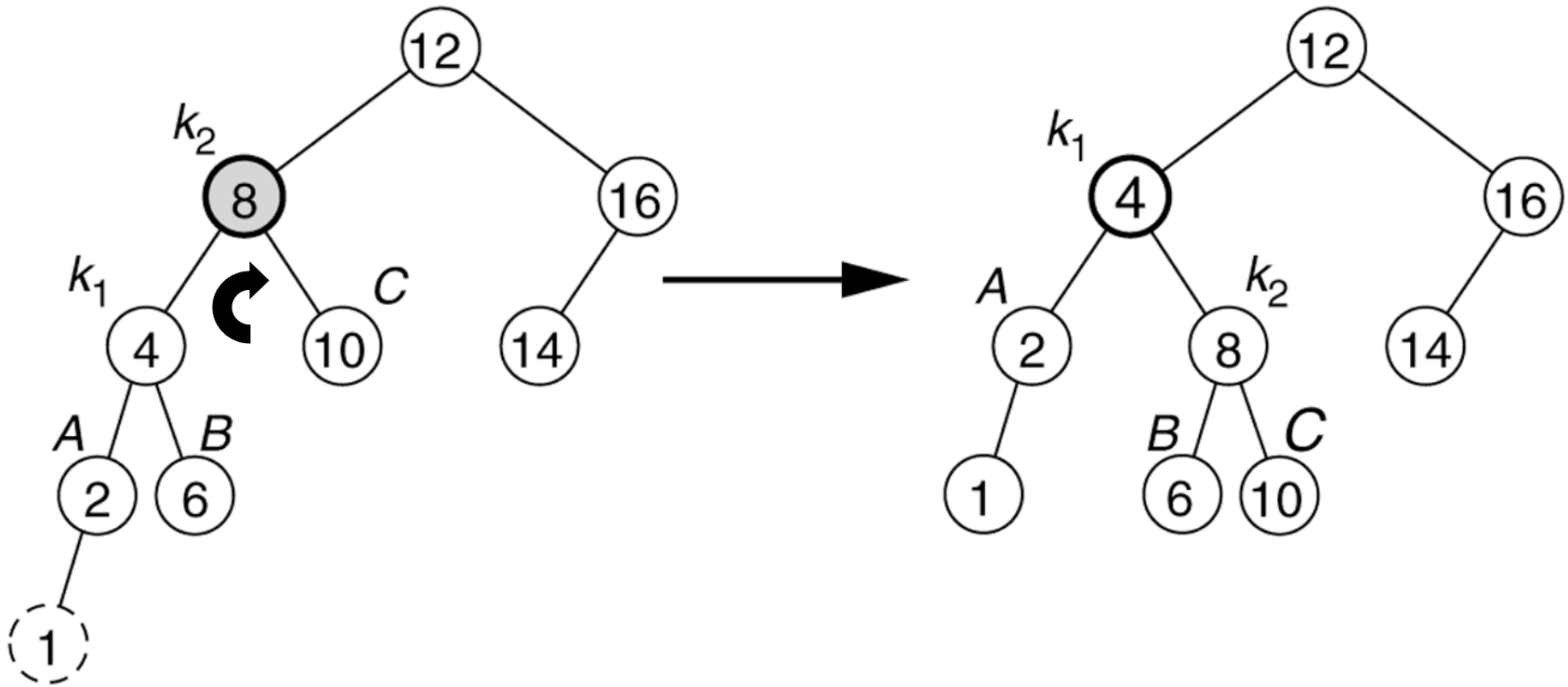
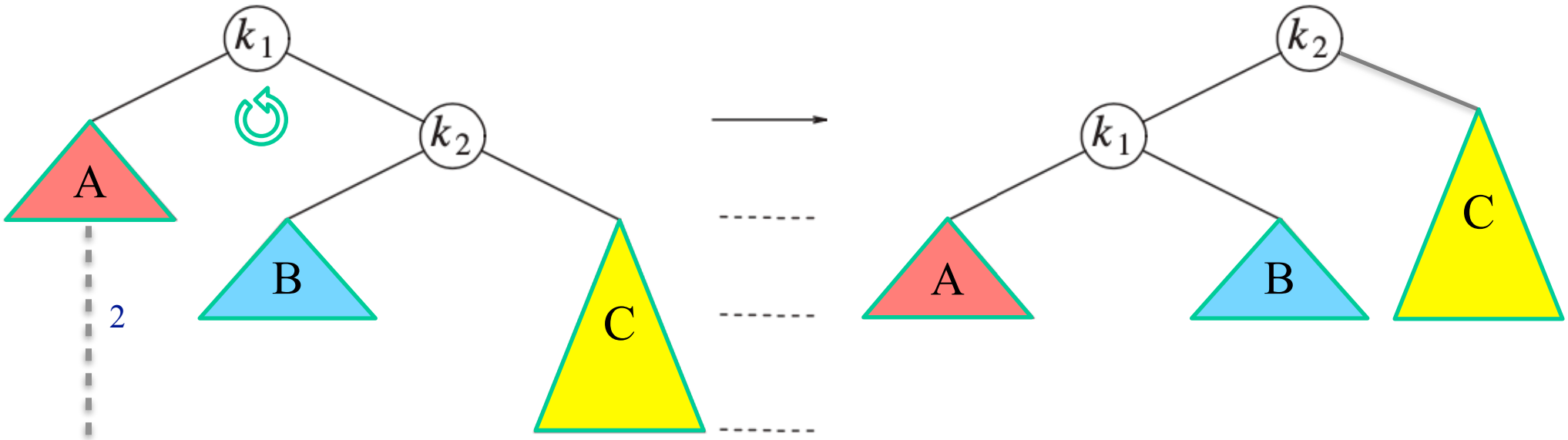
Single Right Rotation – Example



(a) Before rotation

(b) After rotation

Single left rotation to fix case 4



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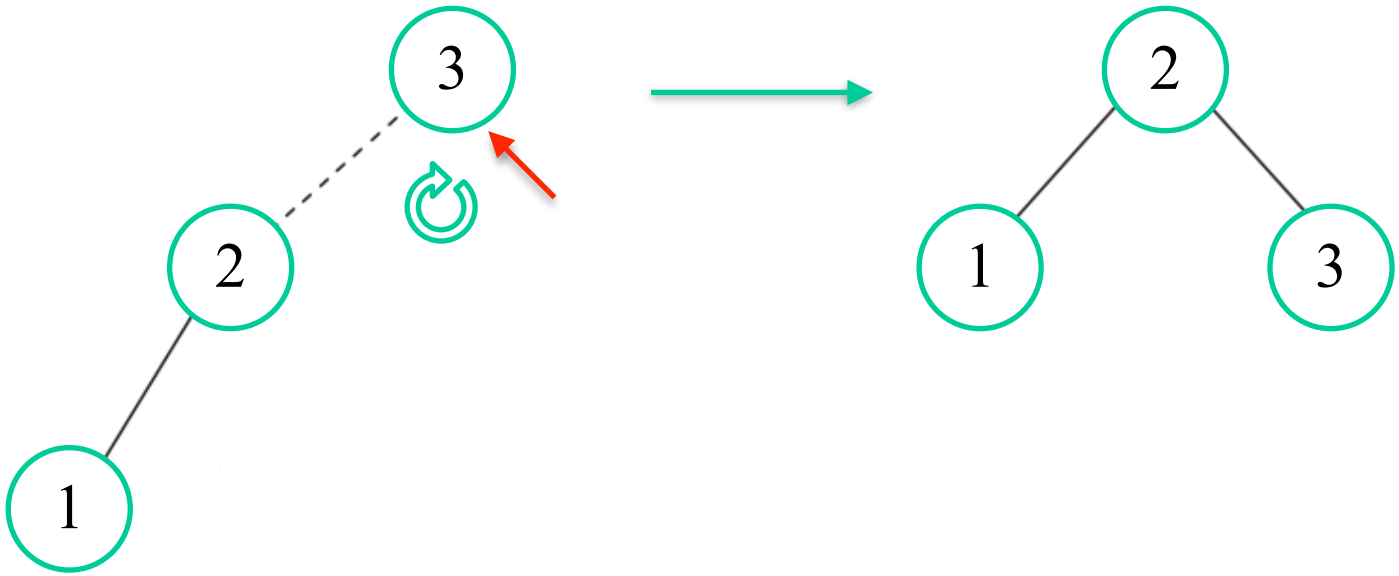
Example: insert the items 3, 2, 1, and then 4 through 7 in sequential order



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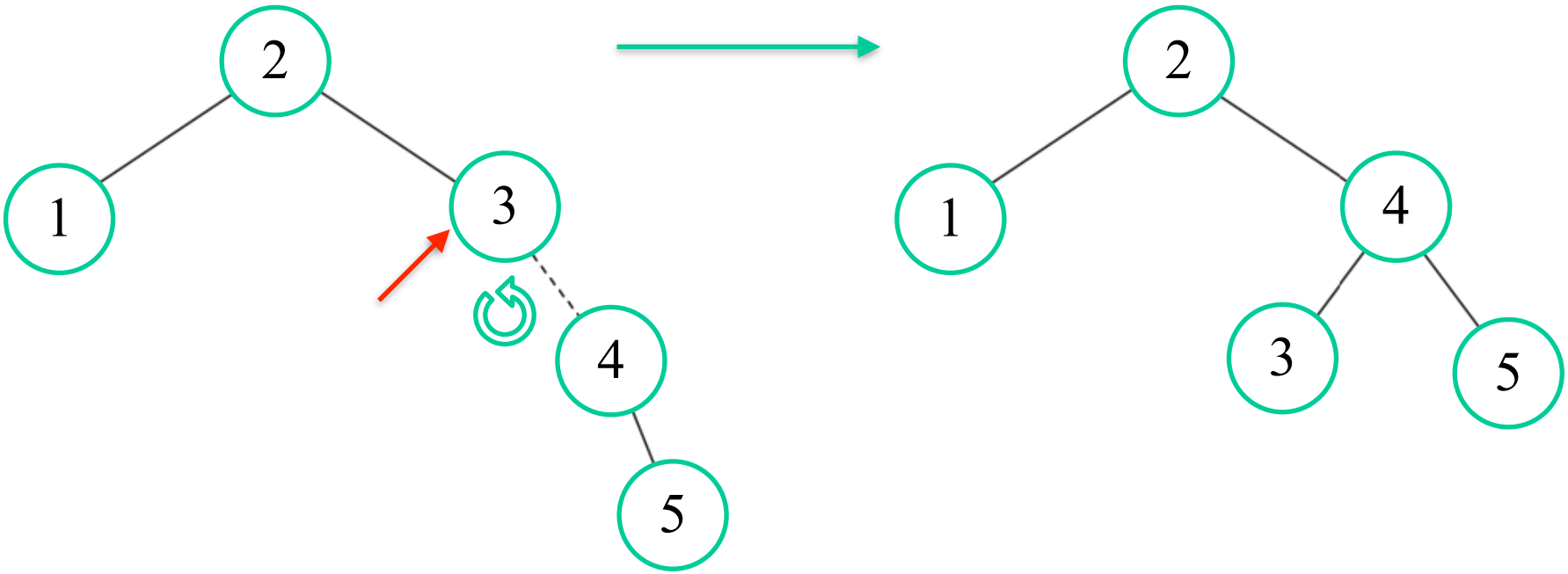

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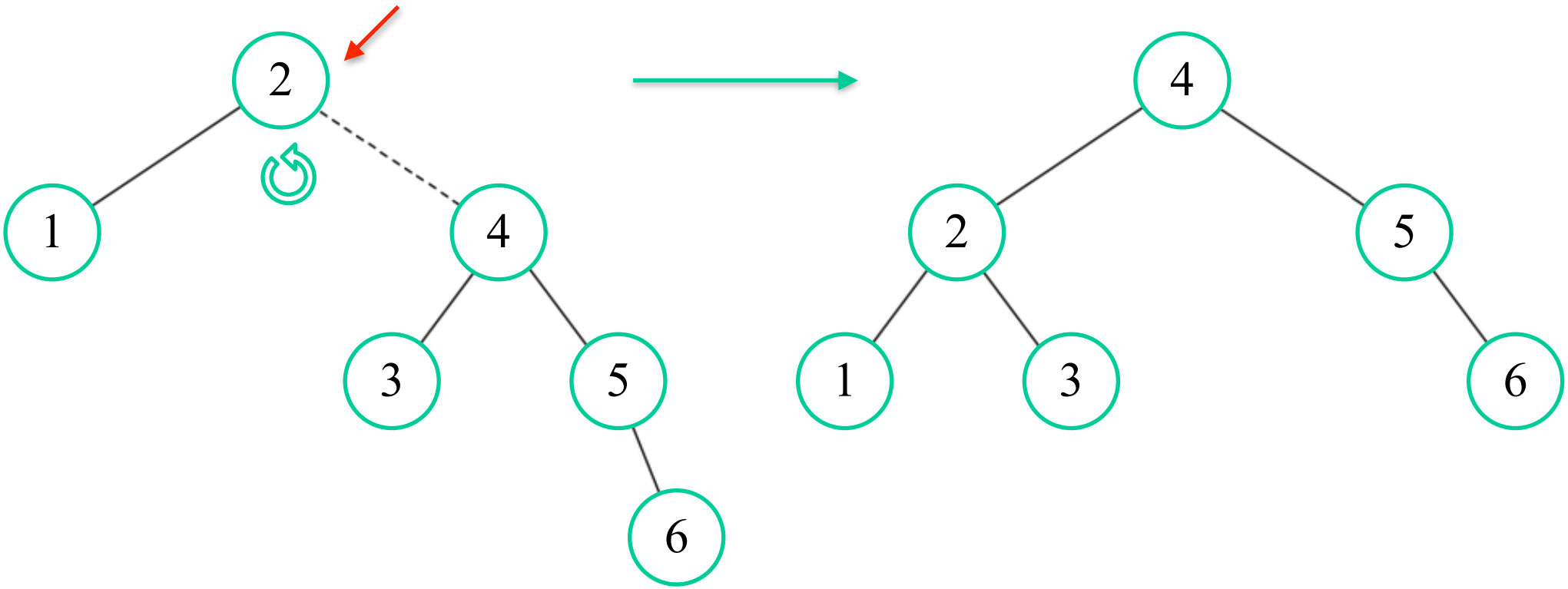

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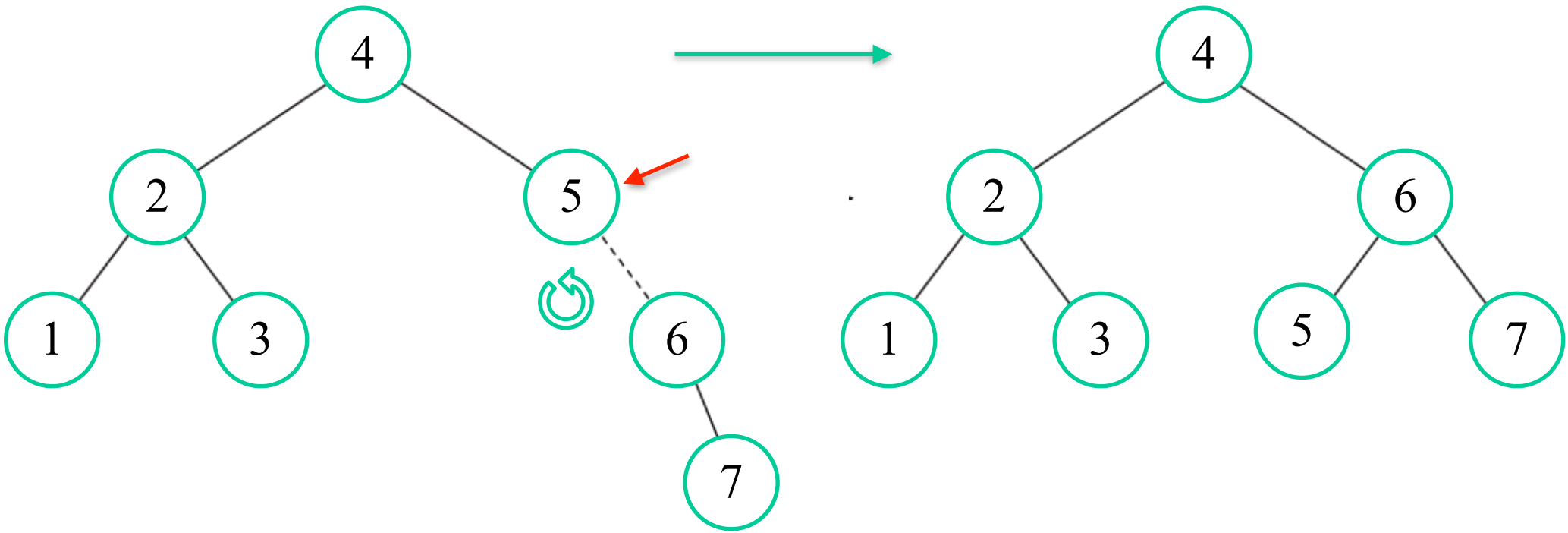

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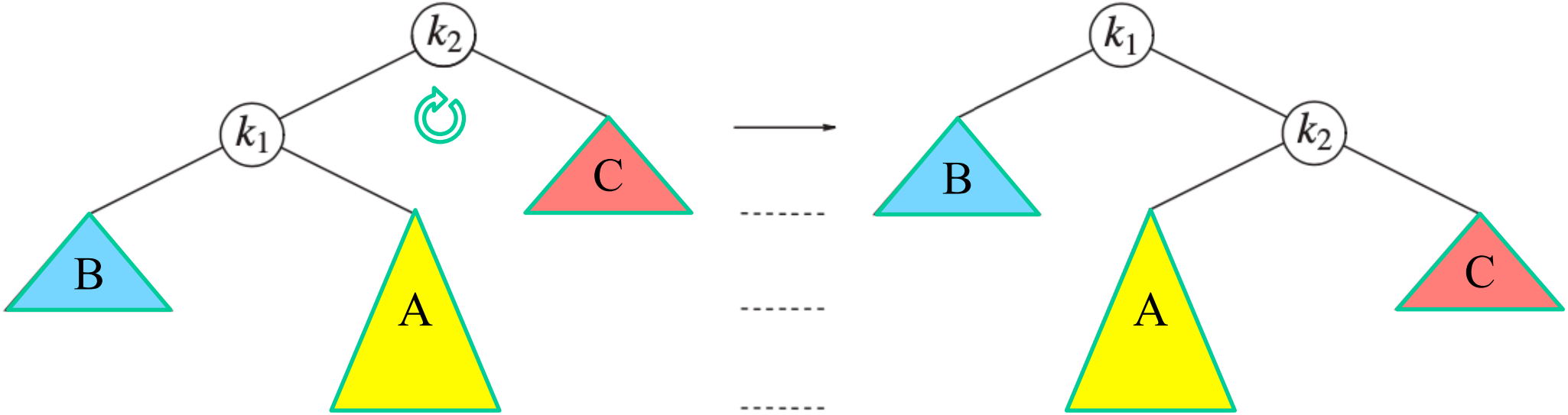
Remember Case 2: An insertion into the right subtree of the left child of x

Single rotation fails to fix case 2

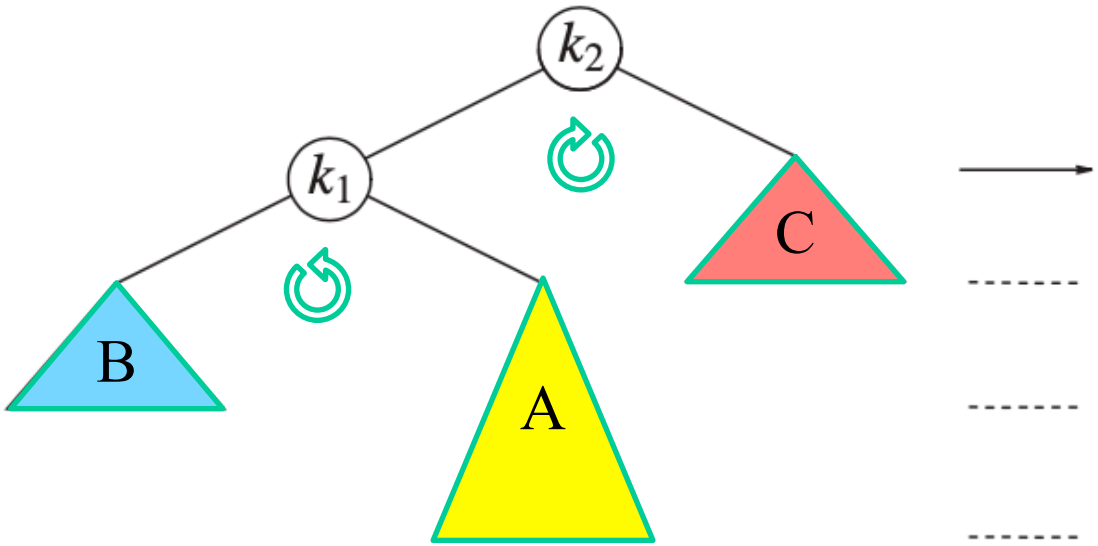


Problem: Subtree *A* is too deep, and a single rotation is **not** a solution

Remember Case 2: An insertion into the right subtree of the left child of x

Single rotation fails to fix case 2

Double rotation solves the case



Next Slide

Problem: Subtree *A* is too deep, and a single rotation is not a solution

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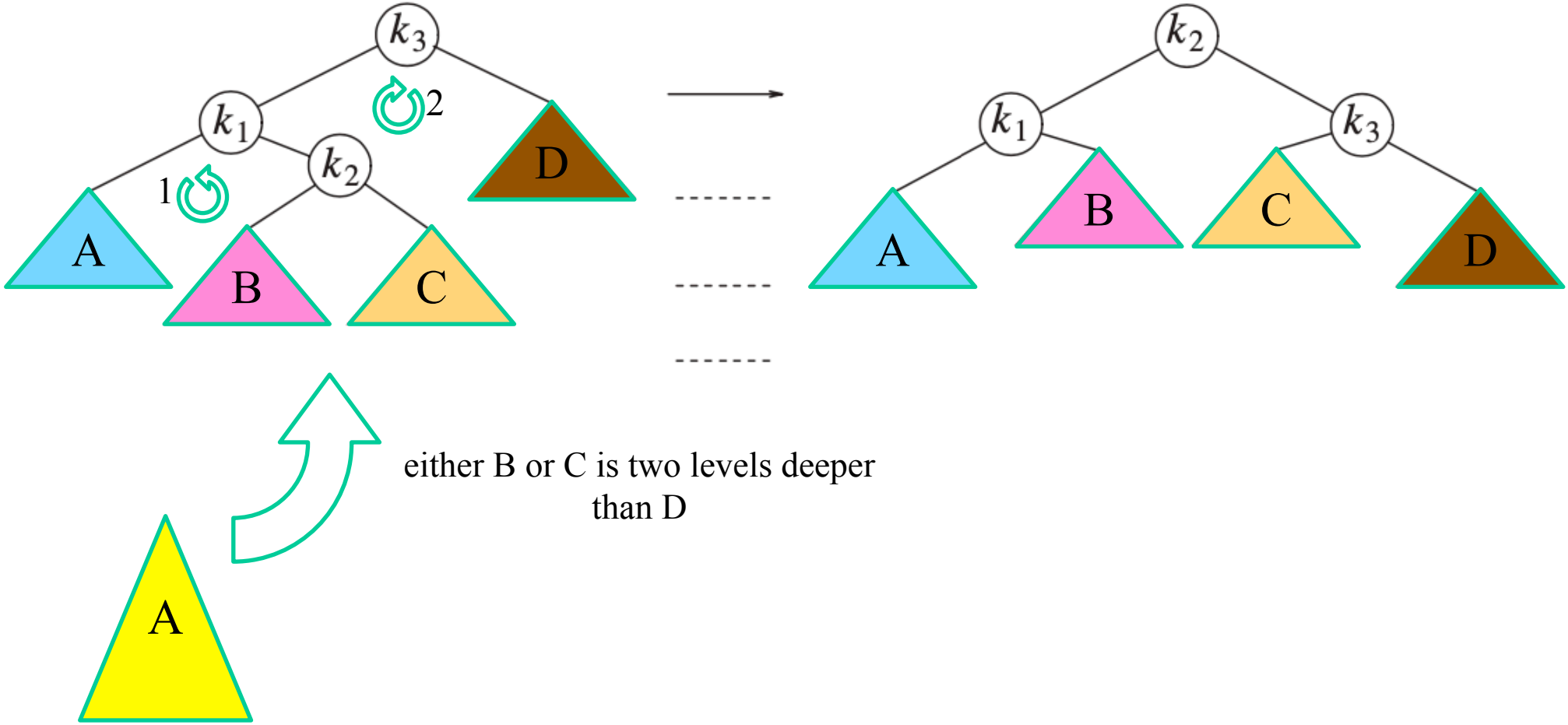
Remember Case 2: An insertion into the right subtree of the left child of x

Left-right double rotation



either B or C is two levels deeper than D

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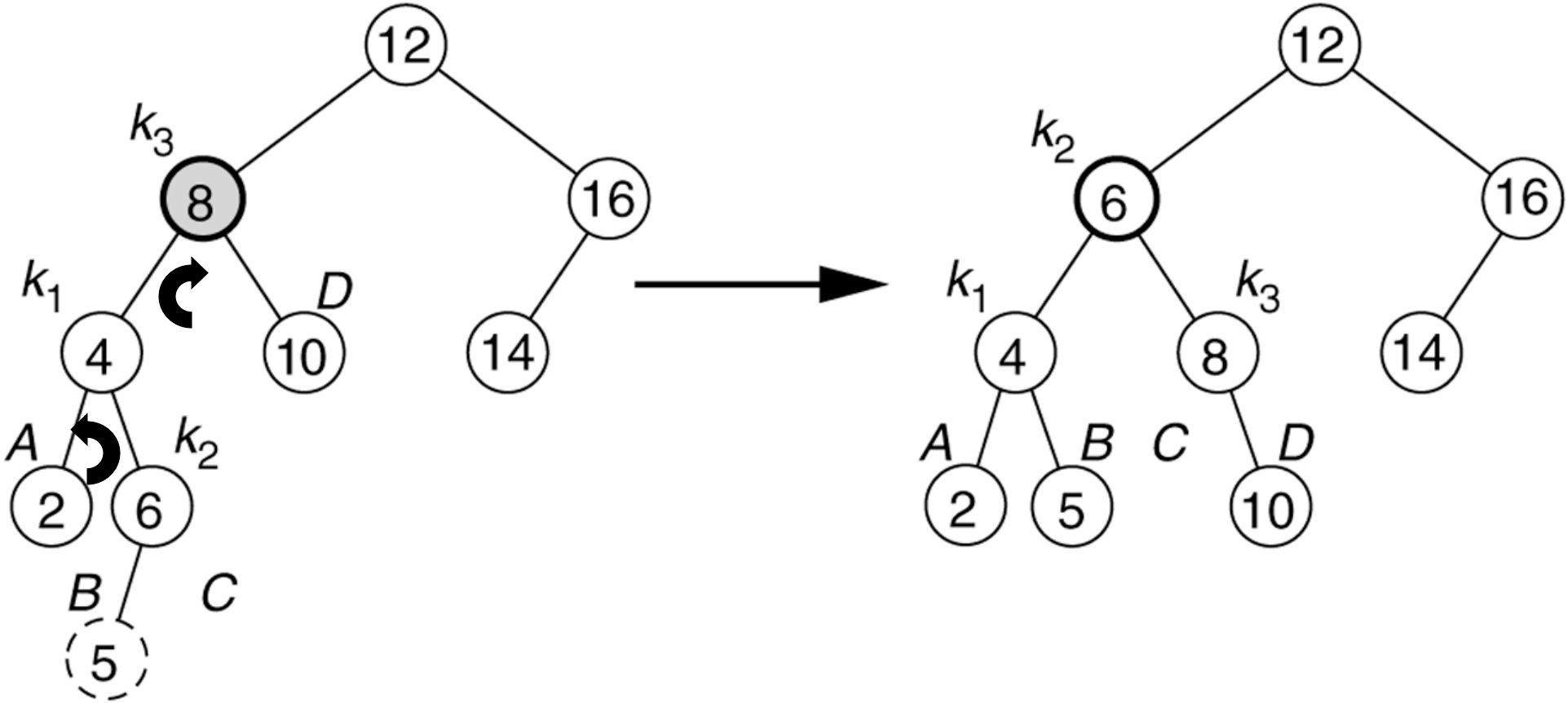
Left-Right Double Rotation – Example

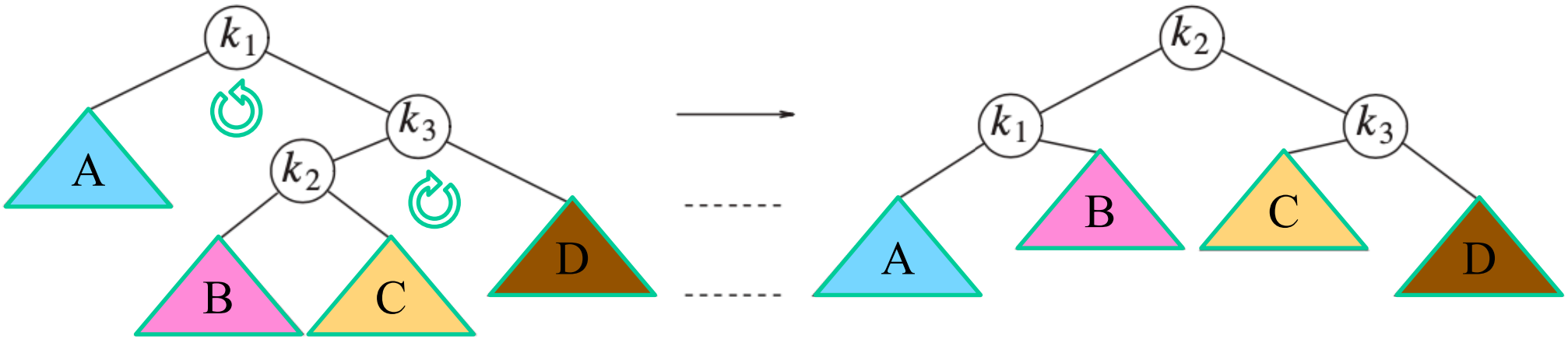


(a) Before rotation

(b) After rotation

Remember Case 3: An insertion into the left subtree of the right child of x

Right-left double rotation fixes case 3



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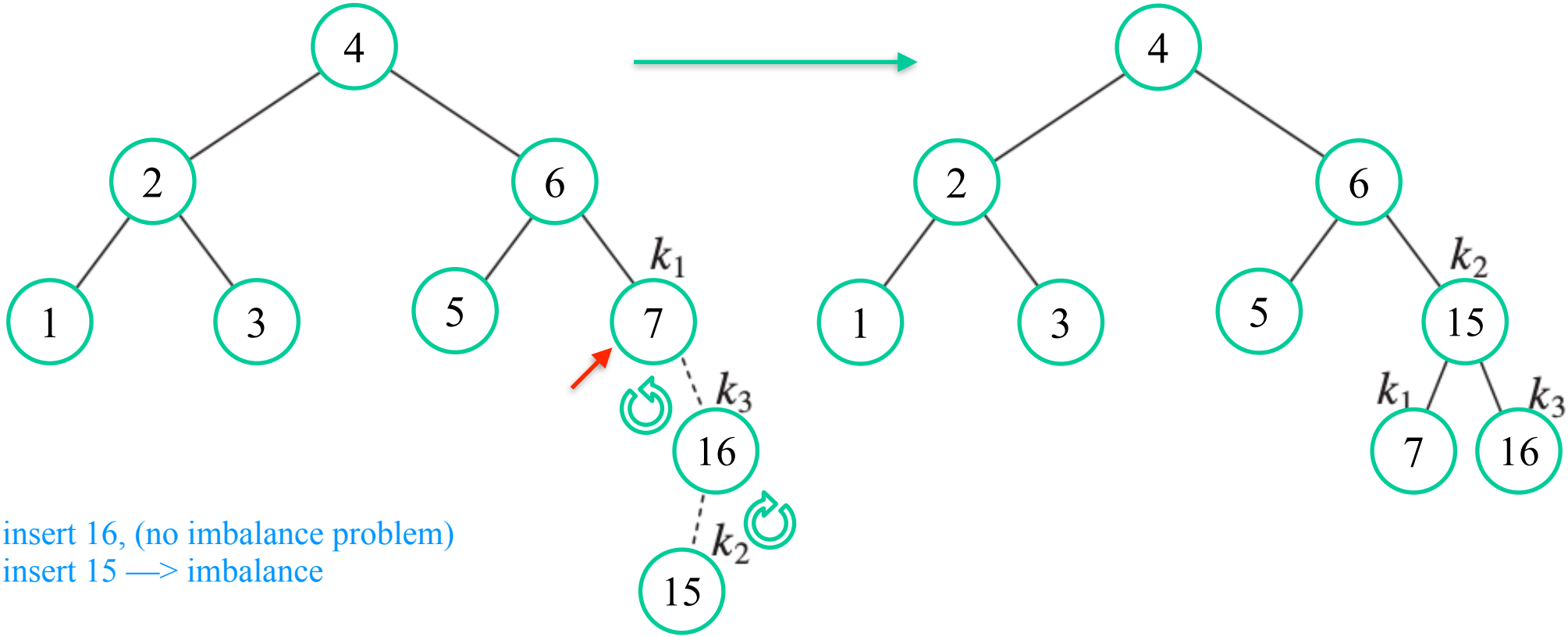
Example: Continue…

insert the items 3, 2, 1, and then 4 through 7 in sequential order, and insert 10 through 16 in reverse order



insert 16, (no imbalance problem)
insert 15 —> imbalance

Right-left double rotation

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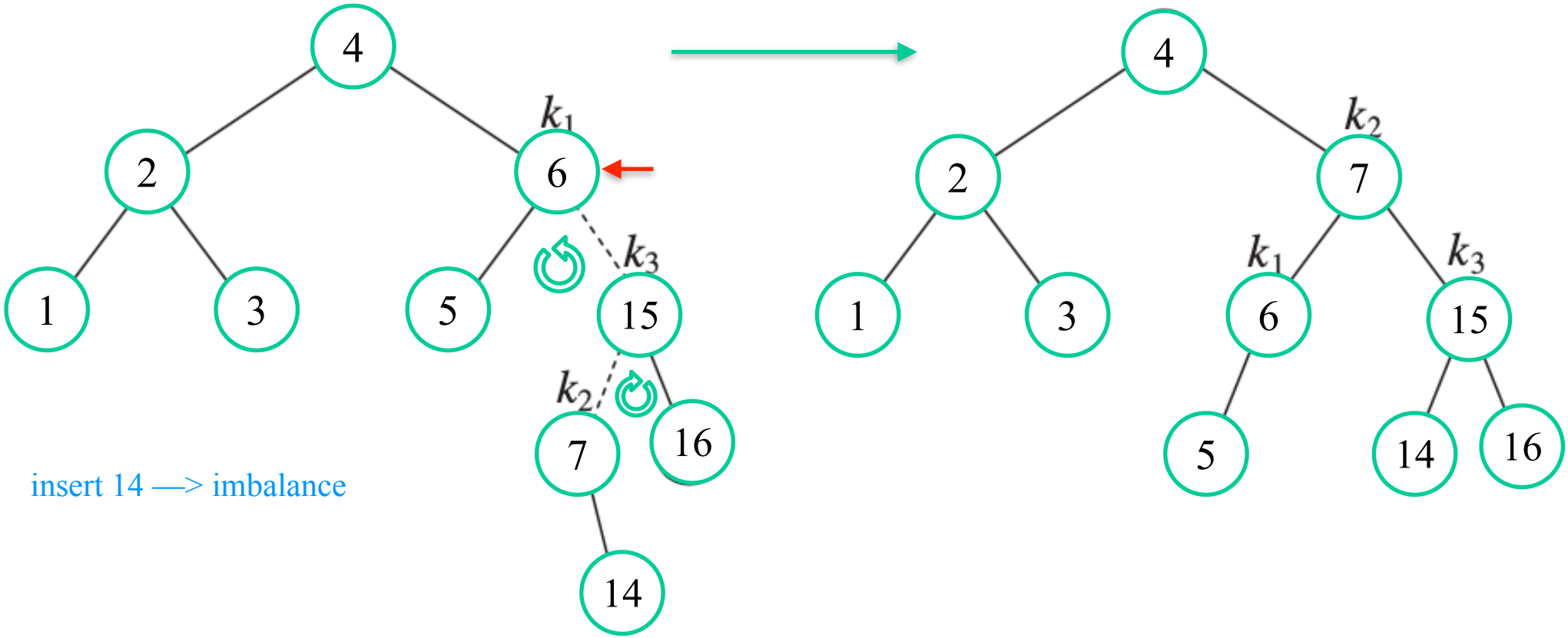
Example: Continue…

insert the items 3, 2, 1, and then 4 through 7 in sequential order, and insert 10 through 16 in reverse order

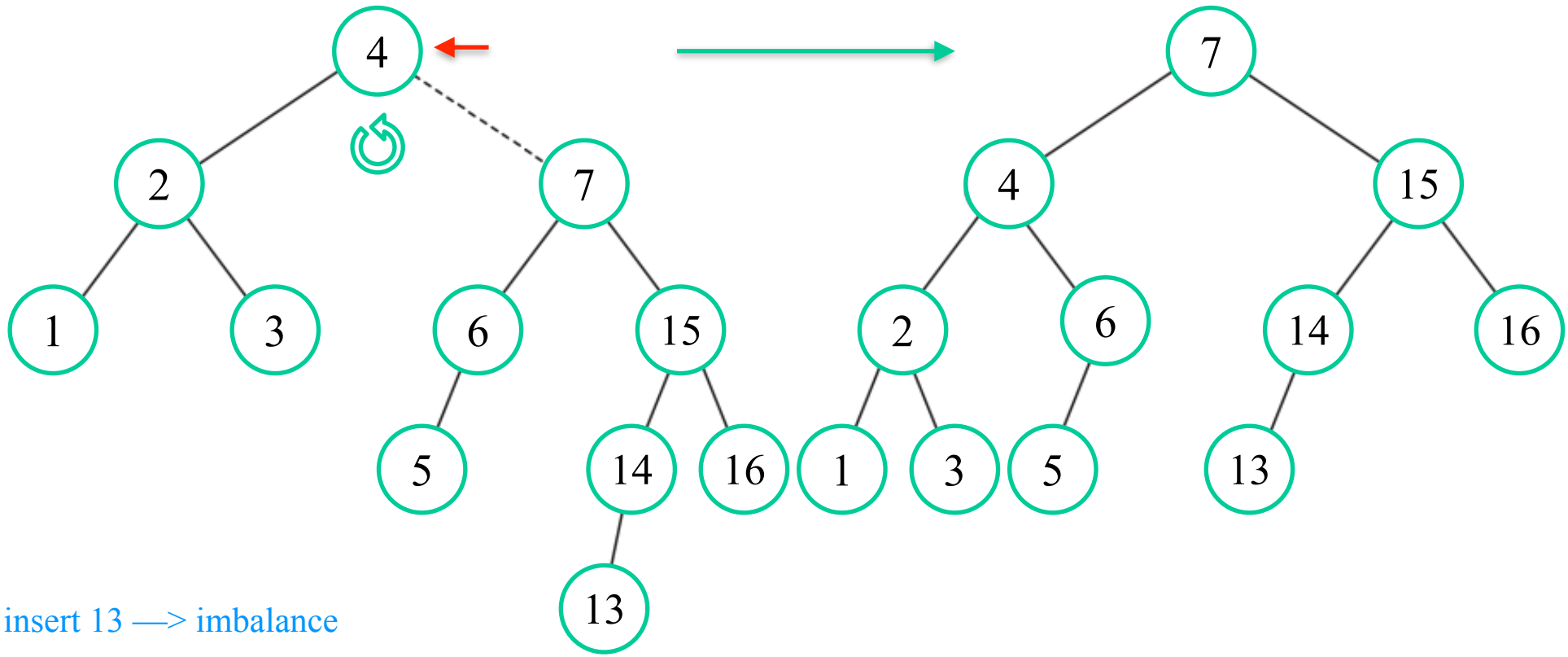


insert 14 —> imbalance

Right-left double rotation

Example: Continue…

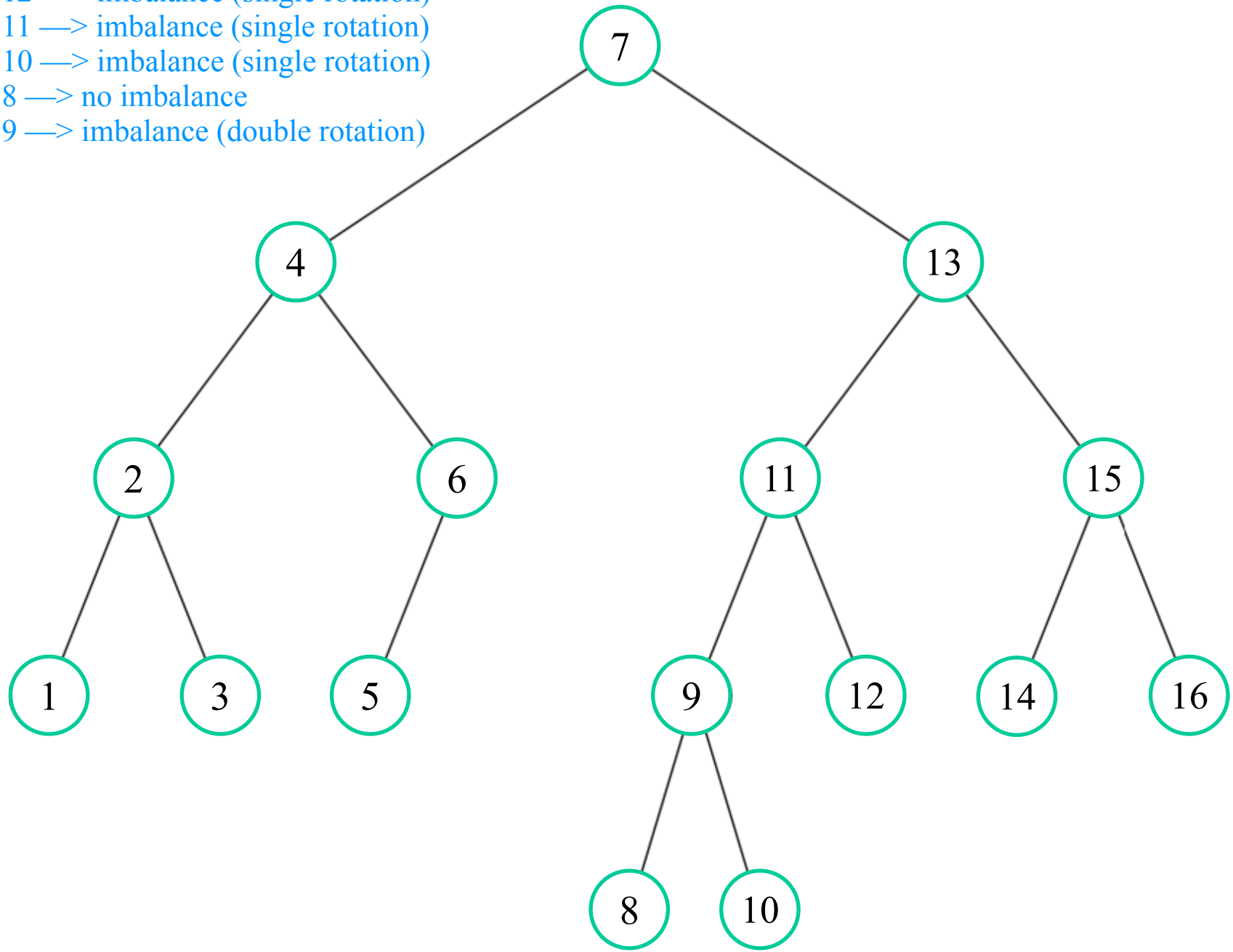
insert the items 3, 2, 1, and then 4 through 7 in sequential order, and insert 10 through 16 in reverse order



insert 13 —> imbalance

Single left rotation

insert 12 —> imbalance (single rotation)
insert 11 —> imbalance (single rotation)
insert 10 —> imbalance (single rotation)
insert 8 —> no imbalance
insert 9 —> imbalance (double rotation)



Insertion – Analysis

- Single rotation preserves the original height:
  - The new height of the entire subtree is exactly the same as the height of the original subtree before the insertion.
- Therefore it is enough to do rotation only at the first node, where imbalance exists, on the path from inserted node to the root.
- Thus the rotation takes O(1) time.
- Hence insertion is O(logN)

AVL Tree -- Deletion

- Deletion is more complicated.
- Deletion of a node x from an AVL tree requires the same basic ideas, including single and double rotations, that are used for insertion.
- *We may need more than one rebalance on the path from deleted node to root.*
- Deletion is O(logN)

AVL Tree – Deletion Method

- First delete the node (to be deleted) same as the deletion in a binary search tree.
  - ***Leaf Node :*** Delete the node
  - ***Node with Single Child:*** Delete the node, the parent of the deleted node points the single child of the deleted node after deletion.
  - ***Node with Two Children:*** Find the inorder successor of the node (to be deleted) and swap them. Then delete the inorder successor.

- Walk through from *the deleted node back to the root*, and *rebalance* the nodes on the path if it is required.
- We'll *trace* the effects of this deletion *on height through all the nodes on the path from the deleted node x back to the root*.

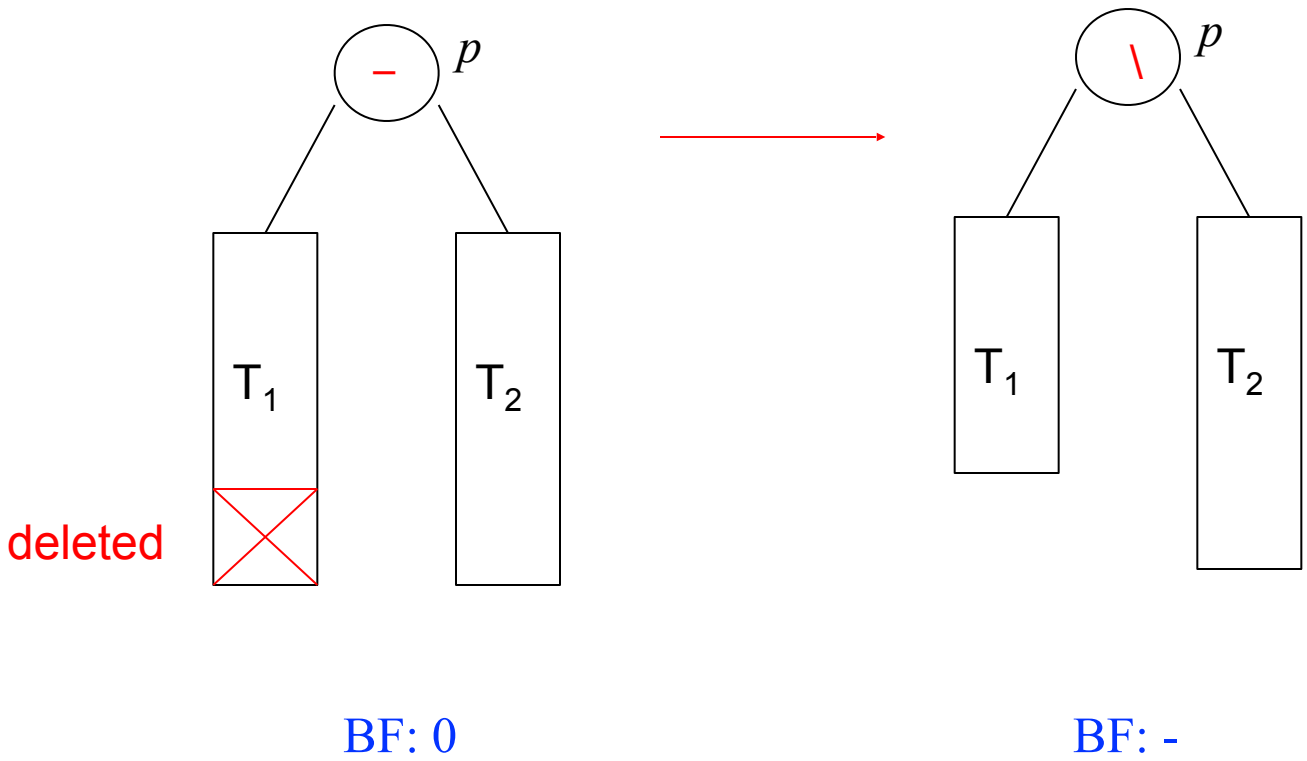
AVL Tree – Deletion Method (cont.)

- A boolean variable <u>shorter</u> shows if the height of a subtree has been shortened.
- Each node of the AVL tree is associated a ***balance factor.***
  - ***left high*** -- left subtree has height greater than right subtree (+)
  - ***right high*** -- right subtree has height greater than left subtree (-)
  - ***equal*** -- left and right subtrees have same height (0)
- The action to be taken at each node depends on
  - the value of shorter
  - balance factor of the node
  - sometimes the balance factor of a child of the node.
- shorter is initially *true*.
- The following steps are to be done for each node *p* on the path from the parent of x to the root, *provided shorter remains true*.
- When shorter becomes false, *the algorithm terminates*.

Case 1

Case 1: ***The current node p has balance factor equal.***

- Change the balance factor of *p* to *right-high*.
- shorter becomes false



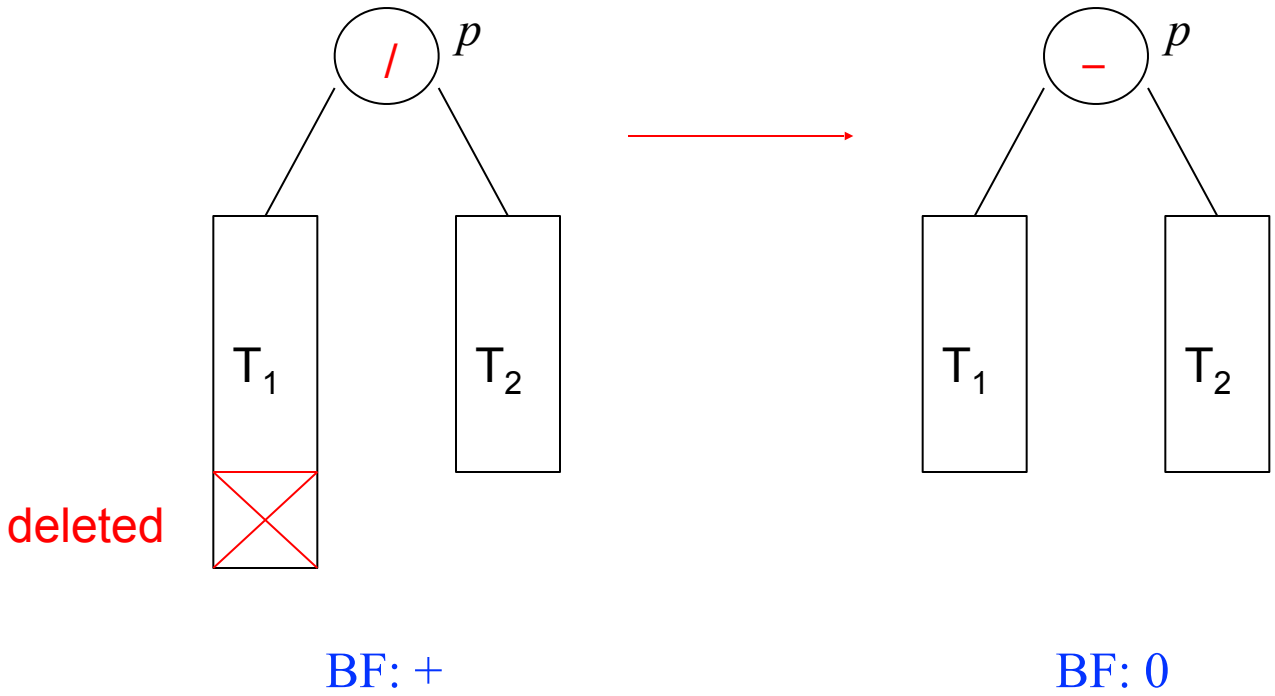
- No rotations
- Height unchanged

BF: Left Height - Right Height

Case 2

Case 2: The balance factor of p is not equal, and the taller subtree was shortened.

- Change the balance factor of *p* to equal
- Leave shorter true.



- No rotations
- Height reduced

Case 3

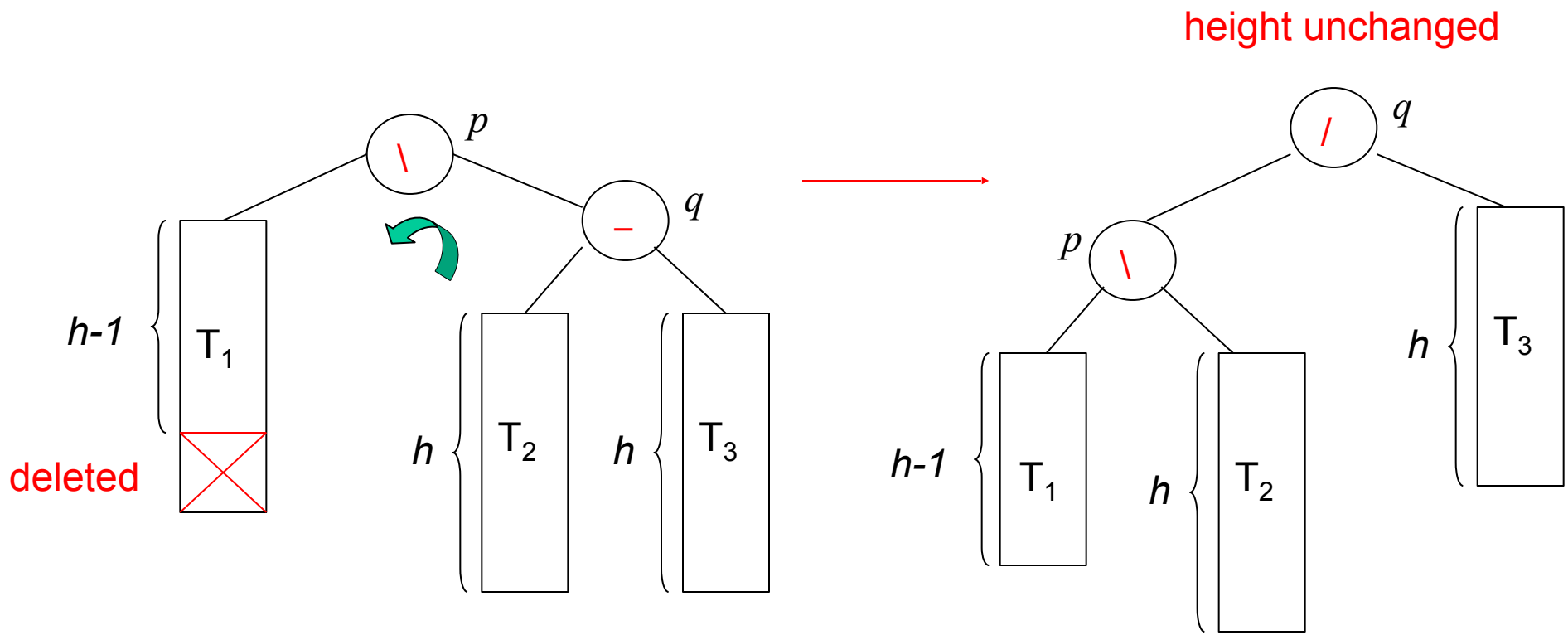
Case* 3: *The balance factor of p is not equal,
and the shorter subtree was shortened.

- Rotation is needed.
- Let q be the root of the taller subtree of p.
- We have *three sub-cases* according to the balance factor of q.

Case 3a

Case 3a*: *The balance factor of q is equal.

- Apply a single rotation
- shorter becomes false.

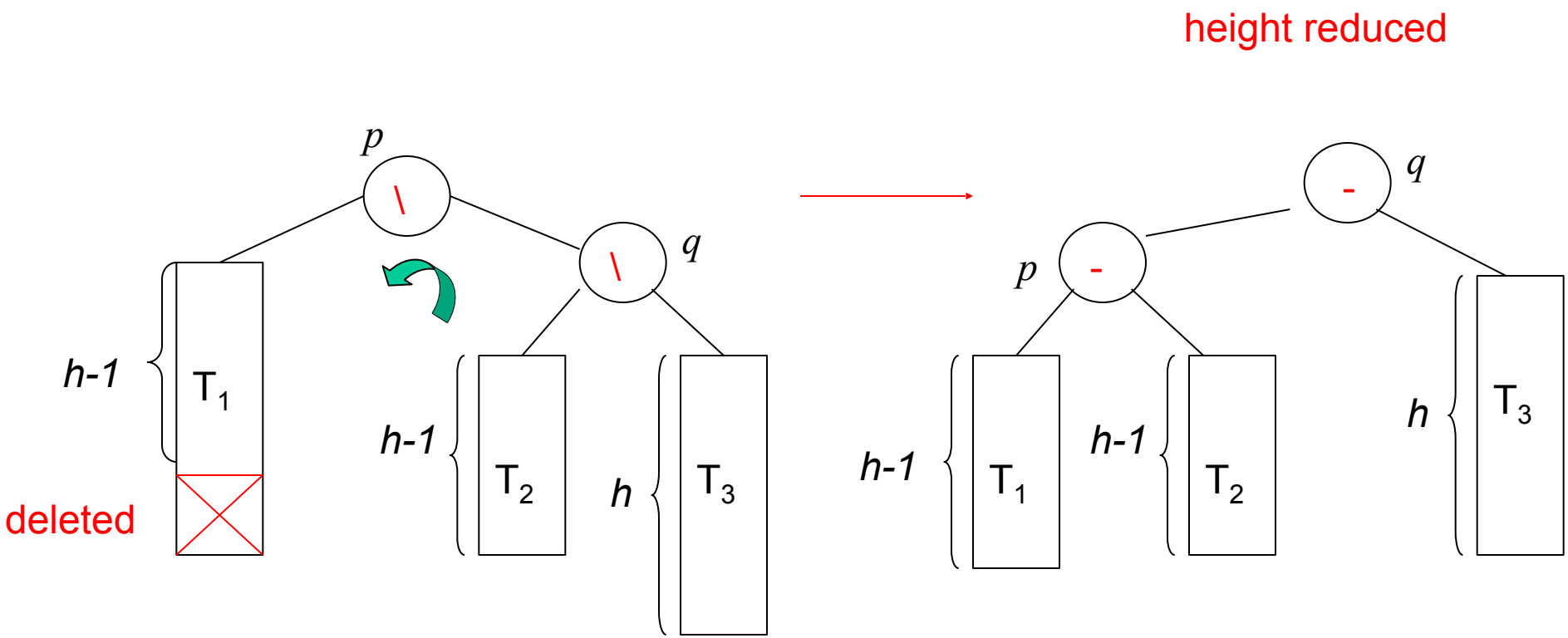


BF(p): - BF(q): 0

Case 3b

Case 3b*: *The balance factor of q is the same as that of p.

- Apply a single rotation
- Set the balance factors of p and q to equal
- leave shorter as true.

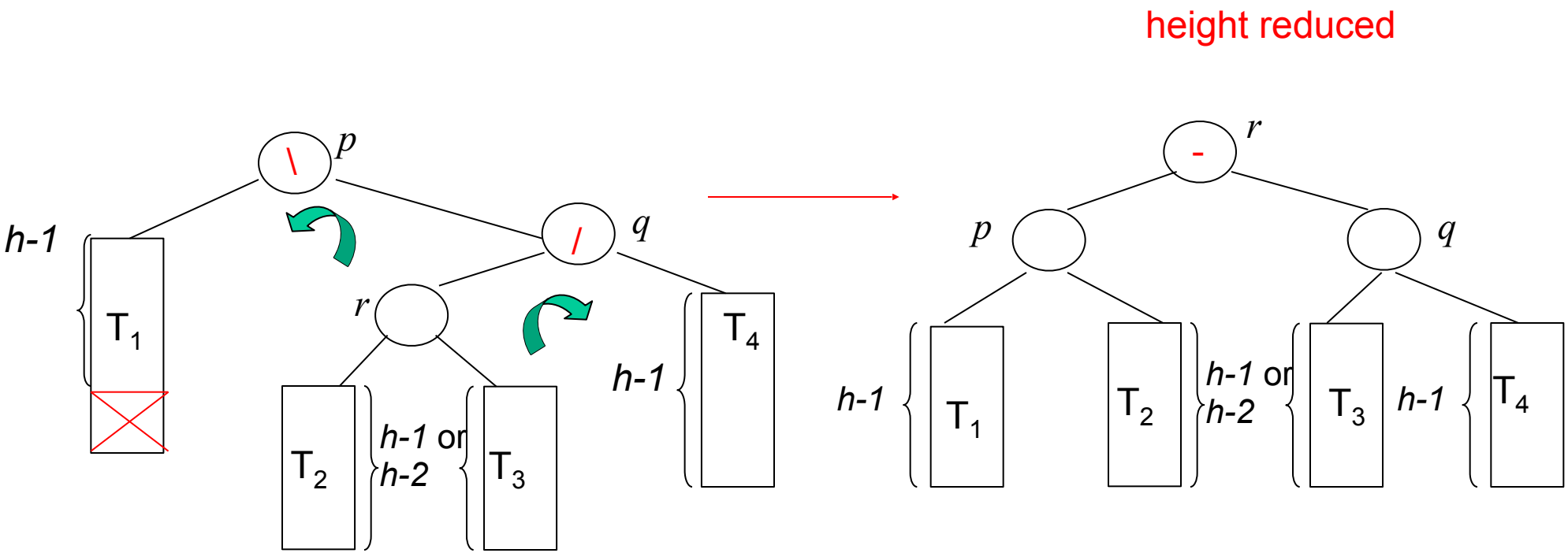


BF(p): - BF(q): -

Case 3c

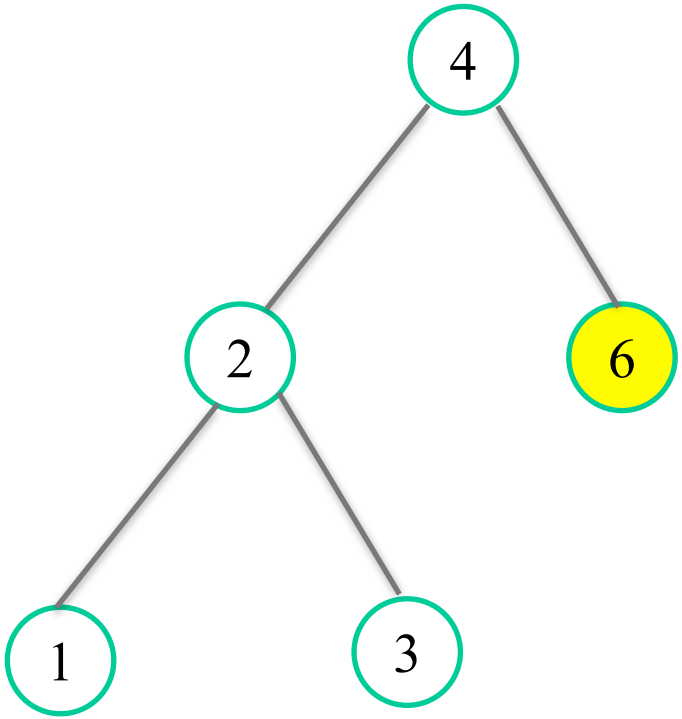
Case 3c: The balance factors of p and q are opposite.

- Apply a double rotation
- set the balance factors of the new root to equal
- leave shorter as true.

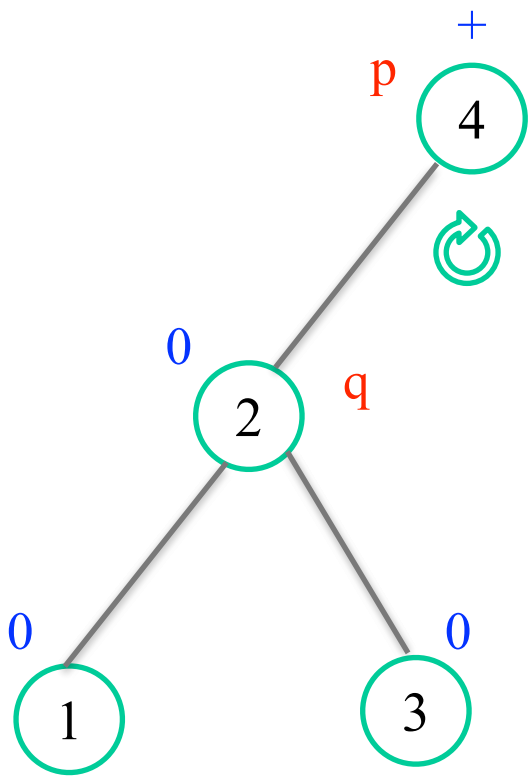


BF(p): - BF(q): +

Delete Node 6

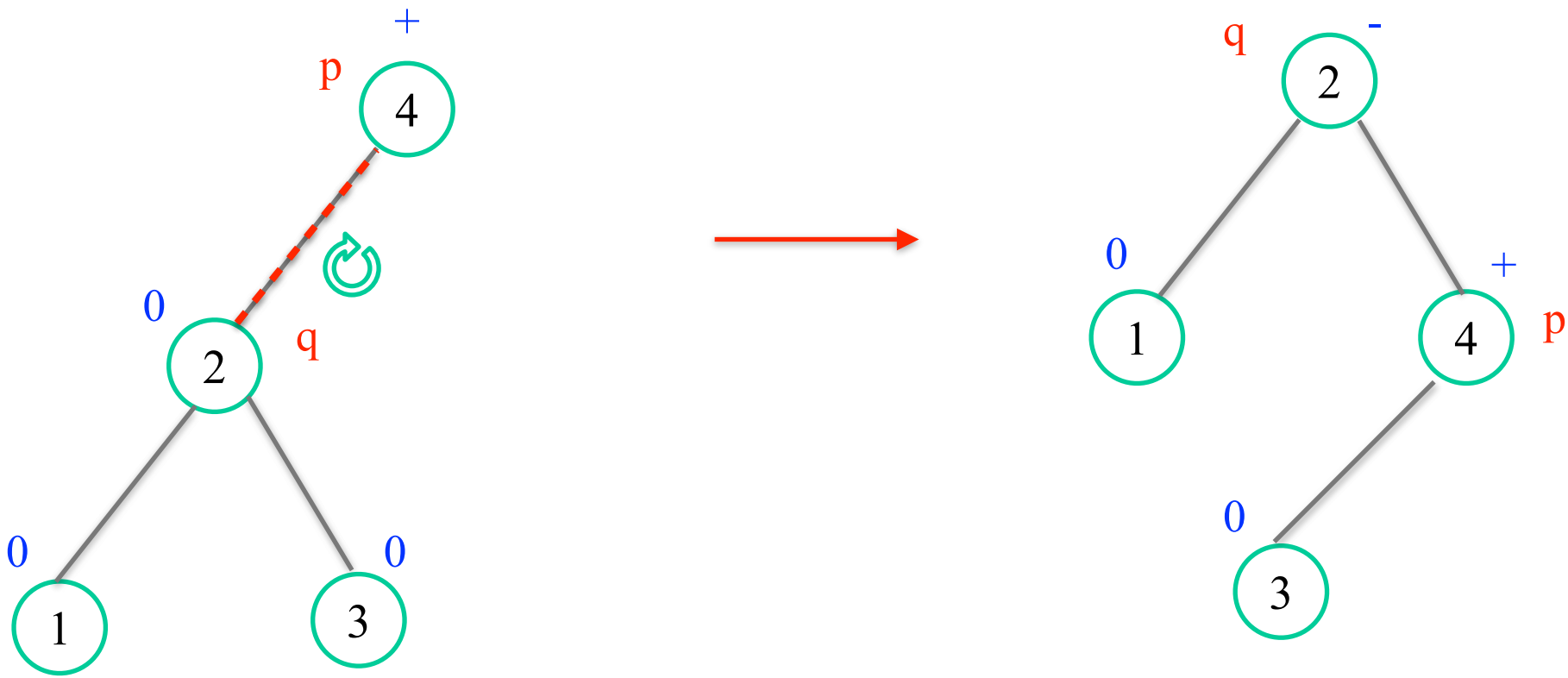


Delete Node 6



Requires rotation
Balance factor of q is equal to 0
Case 3a

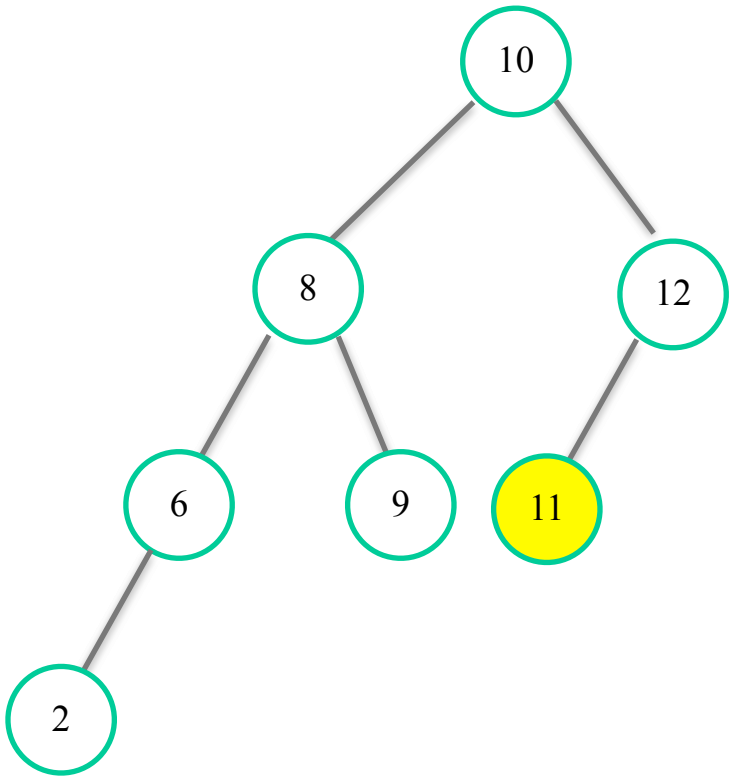
Delete Node 6



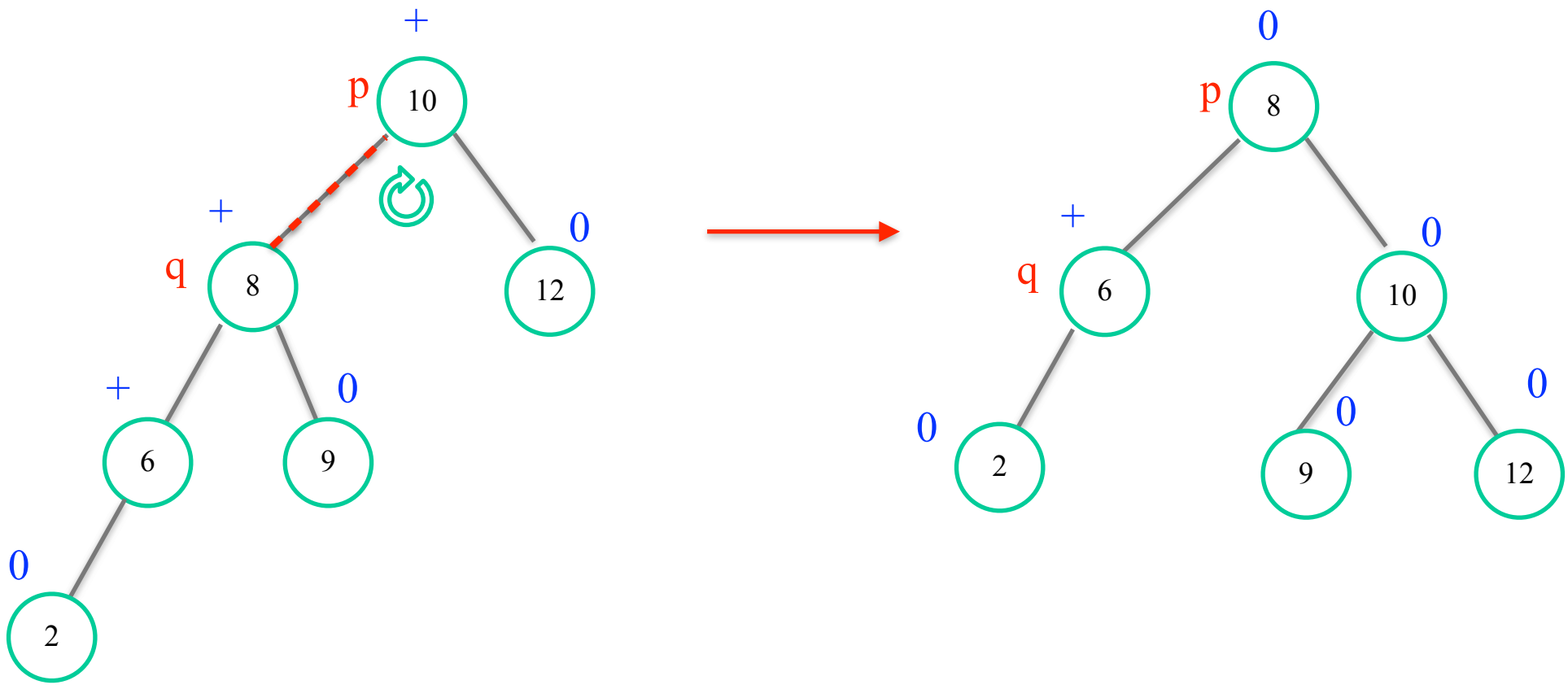
Requires rotation
Balance factor of q is equal to 0
Case 3a

Height unchanged

Delete Node 11



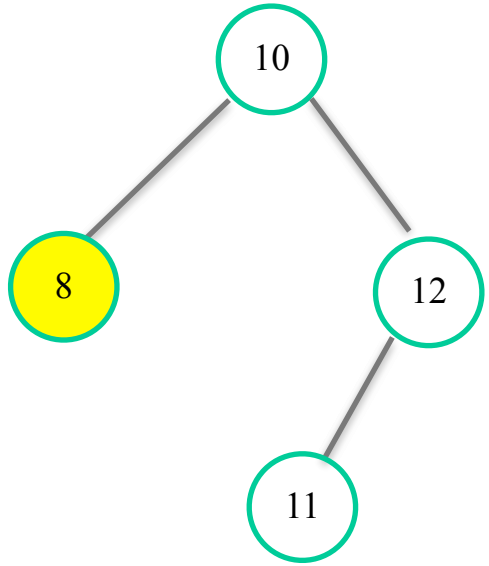
Delete Node 11

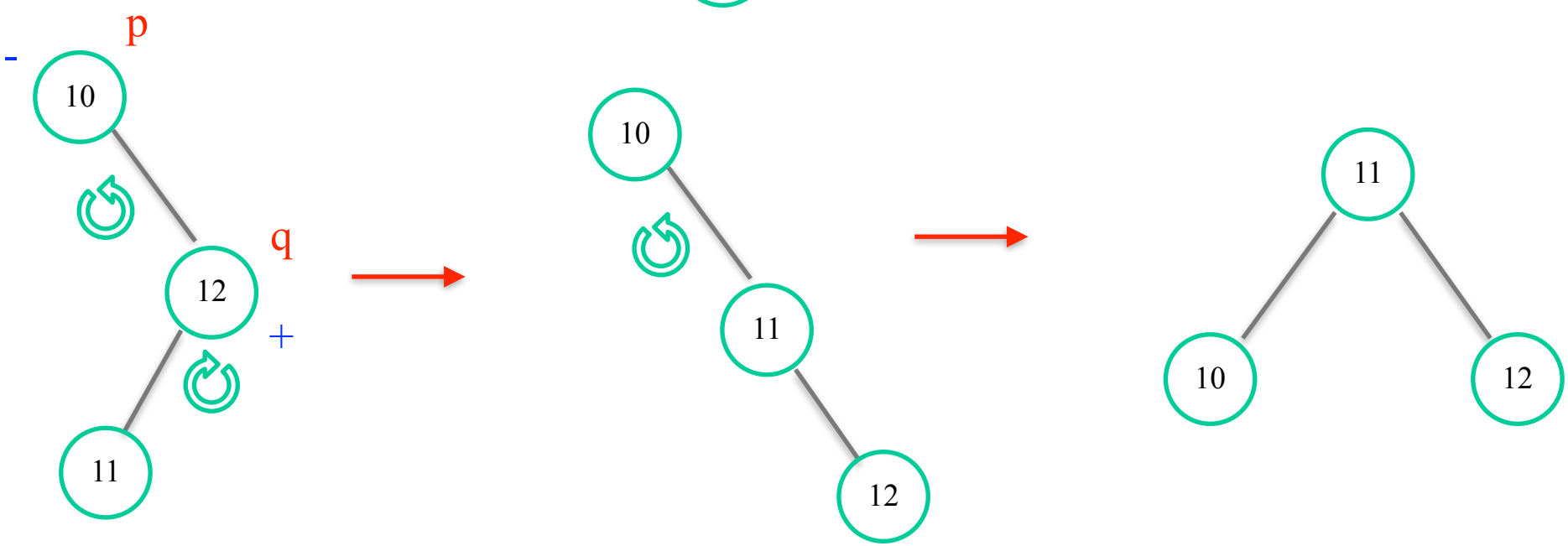


Balance factors of p and q are the same
Case 3b

Balance factors of
p and q are opposite
Case 3c

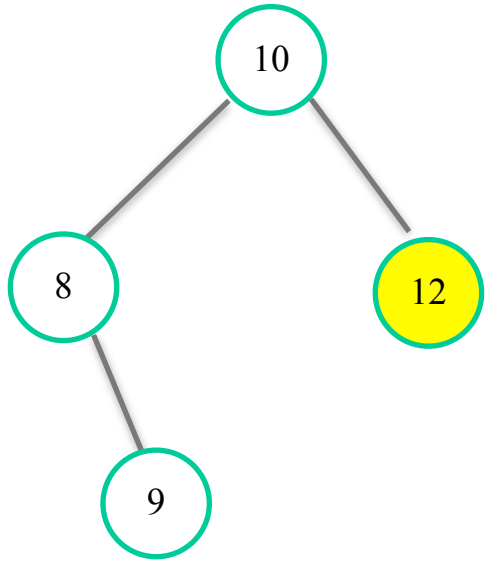
Delete Node 8

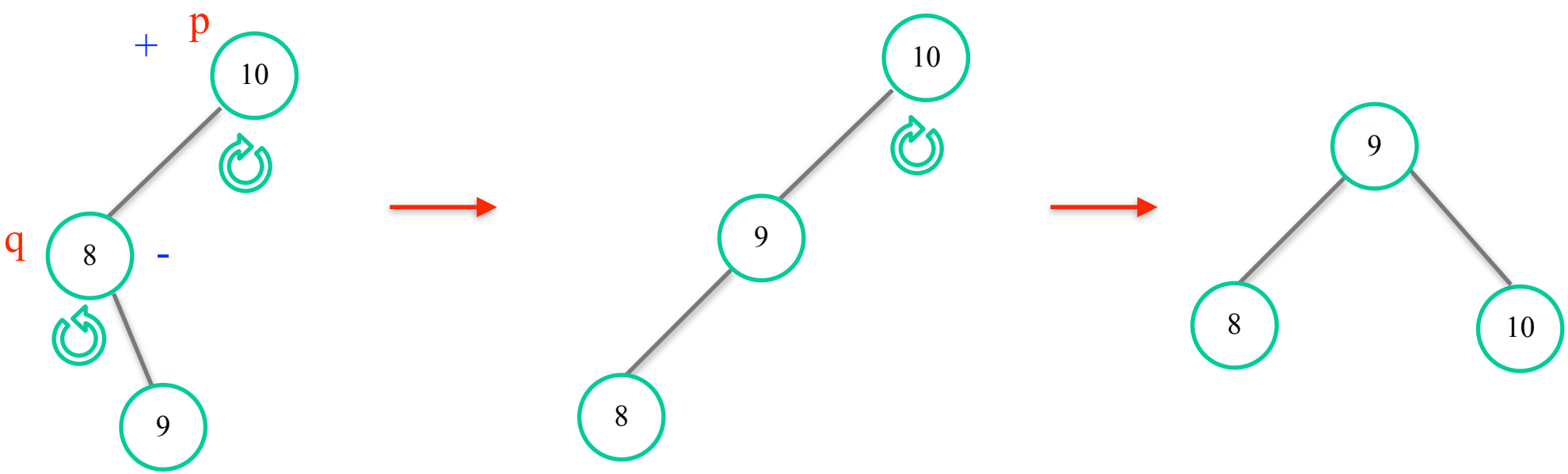




Balance factors of
p and q are opposite
Case 3c

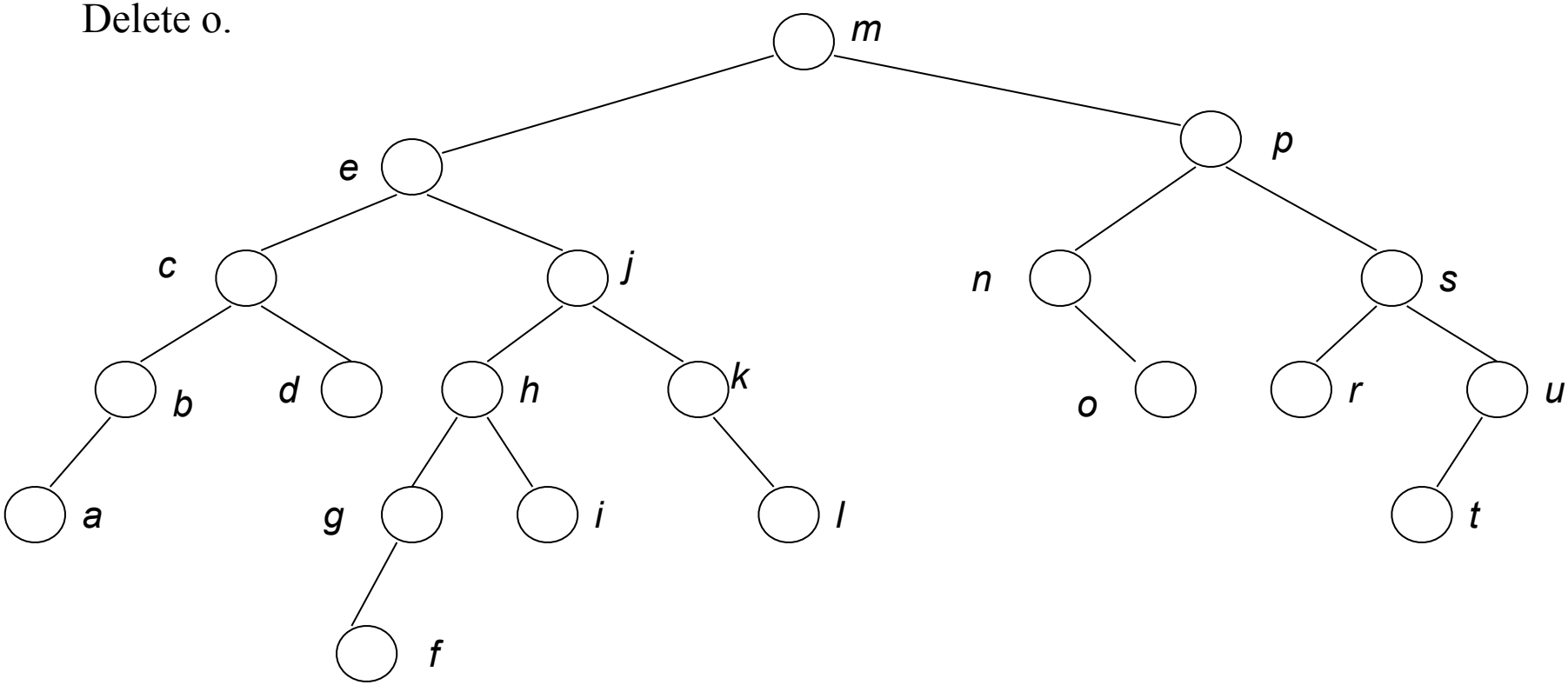
Delete Node 12





Example

Delete o.



Worst Cases of AVL Trees

H	minN	logN	H / logN
4	7	2,81	1,42
5	12	3,58	1,39
6	20	4,32	1,39
7	33	5,04	1,39
8	54	5,75	1,39
9	88	6,46	1,39
10	143	7,16	1,40
11	232	7,86	1,40
12	376	8,55	1,40
13	609	9,25	1,41
14	986	9,95	1,41
15	1.596	10,64	1,41
16	2.583	11,33	1,41
17	4.180	12,03	1,41
18	6.764	12,72	1,41
19	10.945	13,42	1,42
20	17.710	14,11	1,42
30	**2.178.308**	**21,05**	**1,42**
40	**267.914.295**	**28,00**	**1,43**
50	**32.951.280.098**	**34,94**	**1,43**

What is the minimum number of nodes in a height N AVL tree?

minN(0) = 0
minN(1) = 1
minN(2) = 2
minN(3) = 4
.

minN(h) = minN(h-1) + minN(h-2) + 1

Maximum height of a N-node AVL tree is less than 1.44 logN